Patrick Thangarajah

INJURY BIOMECHANICS | OCCUPANT SAFETY | CRASH & IMPACT SIMULATION

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Summary

Mechanical Engineering MASc graduate specializing in injury biomechanics, with experience evaluating injury mechanisms through finite element simulation, physical test validation, injury criteria, and peer-reviewed scientific literature.

Skills

Injury Biomechanics	HIC15, Nij, chest deflection, AIS injury risk, injury-criteria interpretation, THOR-5F, Hybrid III, GHBMC
Simulation and Data Analysis	LS-DYNA, LS-PrePost, FEA, FE model validation, ISO 18571, design of experiments (DOE), JMP
Programming and Engineering Tools	MATLAB, Python, C/C++, LS-PrePost scripting, Microsoft Excel, SolidWorks
Technical Communication	Scientific literature reviews, technical reports, technical presentations, work instructions, SOPs

Education

Master of Applied Science in Mechanical and Mechatronics Engineering — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Research in occupant safety and biomechanics

Bachelor of Applied Science in Mechanical Engineering, Co-op Program — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2018 - May 2023*
- Biomechanics and Medical Devices Option with coursework in human anatomy, physiology, biomaterials, and biomedical design

Work/Research Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Validated a THOR-5F ATD FE model against physical sled testing in two bench configurations in LS-DYNA, achieving an average ISO 18571 score of 0.71 across 10 measurement channels
- Designed and analyzed a 104-simulation DOE-based FE parametric study evaluating the effect of seat belt anchor locations on rear seat occupant response, identifying a rearward and inboard position that reduced peak chest deflection by 16% below the study average
- Communicated findings through a master's thesis, conference paper, technical reports, and six sponsor presentations

Teaching Assistant - MTE 120: Circuits — *Waterloo, ON, Canada*
University of Waterloo — *May 2024 - Aug 2024*
- Supported laboratory instruction for a 100+ student circuits course, assisting students with circuit analysis and measurement, troubleshooting hardware issues, and providing feedback on laboratory work and exams

Summer Research Student — *Vancouver, BC, Canada (Remote)*
Centre for Heart Lung Innovation — *May 2022 - Aug 2022*
- Built a Jupyter Notebook transfer-learning script to diagnose diabetic kidney disease using an emerging imaging platform
- Improved diagnostic accuracy by 10% through model optimization, training refinement, and comparison with existing tissue-slide images

Mechanical Engineering Intern — *Airdrie, AB, Canada*
GN Corporation — *Jan 2022 - Apr 2022*
- Created standardized work instructions and SOPs verified against engineering drawings and customer specifications
- Reduced fiberglass cutting waste by 50% through process analysis, and proposed a material handling workflow projected to halve pick-and-place time

Engineering Intern — *Victoria, BC, Canada*
Breathe Global — *Jul 2021 - Aug 2021*
- Redesigned the analog control circuit for a negative-pressure breathing device in Altium, using LTspice simulation to select low-cost components
- Expanded the device's achievable breathing range from 70% to 88% of target

Projects

Simulation of Cervical Fractures in American Football (ME 720 Course Project) — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Dec 2023*
- Simulated head-first helmet impacts across three velocities (3.4–7.0 m/s) and three neck angles at the lowest speed condition using Hybrid III and GHBMC human body models in LS-DYNA, then evaluated fracture force thresholds and injury location against peer-reviewed literature
- Discovered that predicted peak neck forces (2.26–2.34 kN) fell within the range reported across two cadaveric studies, and that the Neck Injury Criterion exceeded its injury-risk threshold in only 1 of 5 configurations despite fracture occurring in all 5

Soft Distensible Bladder Model - Capstone Project — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2022 - Apr 2023*
- Built a control interface for the soft bladder phantom, enabling real-time pressure/volume data logging, PV/PT curve generation, and automated filling/voiding cycles via four adjustable parameters (max volume, flow rate, jet duration, jet interval)
- Validated pressure chamber performance using the integrated sensor and control system, demonstrating stable, repeatable pressurization at 2.2 kPa (13.8s ramp-up) up to a maximum internal bladder pressure of 4.75 kPa

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- Watson, B., Seeburrun, T., **Thangarajah, P.**, Tang, K., Tylko, S., & Cronin, D. S. (2026). *Experimental and numerical investigation of belt loading for small-stature female occupants in rear seat frontal crash*. Proceedings of the 28th International Technical Conference on the Enhanced Safety of Vehicles (ESV), Toronto, ON, Canada, May 12–15, 2026.
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.